

BUILD NOTES

Aptly

Poora system, shuru se aakhir tak

Ek CV tailoring tool jiski yaaddasht hai. Job post aur CV daalo, exact lines dikhti hain jo badalni hain, ek tap mein lag jaati hain — aur har application ka record rehta hai, taaki paanch hafte baad recruiter call kare toh tumhare paas woh sab ho jo tumne bheja tha.

202

backend tests

7.5k

lines Python

5.2k

lines TypeScript

8

validator layers

5

file formats

~11s

ek tailoring run

CONTENTS

- 01 Ye app hai kya
- 02 Kaise kaam karta hai — step by step
- 03 AI architecture
- 04 No-fabrication validator
- 05 File formats ka jhanjhat
- 06 Tech stack
- 07 Kya bana, kya baaki
- 08 Chalane ka tarika
- 09 Imaandari se

Ye app hai kya

01 THE PROBLEM

Aaj ek job ke liye log **chaar alag tools** use karte hain — CV builder, keyword scanner, application tracker, aur interview prep. Kaam bat jaata hai, AI ka output generic lagta hai, aur sabse stressful moment pe koi tool madad nahi karta.

Woh moment ye hai:

JISKE LIYE YE POORA PRODUCT HAI

Recruiter call karta hai. Tumne paanch hafte pehle apply kiya tha. Yaad nahi kaunsa CV bheja tha, aur job post site se hat chuka hai.

Aptly teen kaam karta hai: **tailor** (exact changes dikhata hai), **remember** (job post ko us din ki haalat mein freeze karke rakhta hai), aur **be ready** (call se pehle sab saamne rakhta hai).

Aur ek rule hai jo kabhi nahi tootta — **ye tumhare baare mein kuch invent nahi karta**. Ye product ka pehla claim bhi hai aur sabse mushkil engineering bhi.

Kaise kaam karta hai

02 STEP BY STEP

Jab tum CV aur job post daalte ho, andar ye hota hai. Har step apne upar wale ka output khaata hai, isliye ye asli sequence hai.

1 File padhi jaati hai

.docx / .pdf / .tex / .txt — har format ka apna parser hai. Ye file ko **CVDocument** mein badalta hai: sections, entries, aur har line ek addressable node.
`backend/src/apty/ingest/`

2 Har line ko ek stable ID milti hai

Yahi cheez one-tap apply possible banati hai. Model kabhi nahi kehta “ye string dhoondh ke badal do” — woh node ID bolta hai. Aur har node apna **SourceAnchor** rakhta hai: asli file mein wapass likhne ka exact pata.
`model/document.py · model/anchors.py`

3 Job post parse hota hai

Ek sasta model advert se requirements, responsibilities aur keywords nikaalta hai. Sirf wahi jo likha hai — salary guess nahi karta, company domain se infer nahi karta.
`gemini-3.1-flash-lite · 3s`

4 Coverage nikalti hai

Job ke har term ke liye check hota hai ki CV usko genuinely dikhati hai ya nahi. “K8s” se “Kubernetes” cover hota hai. Kam score aana *sahi jawab* hai, failure nahi.
`2 / 11 aana bilkul normal hai`

5 Har section parallel mein tailor hota hai

Summary, experience, projects, skills — sab alag-alag calls, ek saath. Isliye pehla change card ~7 second mein aa jaata hai, poora run khatam hone ka wait nahi karna padta.
`gemini-2.5-flash · parallel`

6 Validator har suggestion ko rokta hai

Model ka output seedha screen pe nahi jaata. 8 layers ka deterministic Python check chalta hai. Jo pass nahi hota woh discard — aur count tumhe dikhati hai.
`validate/__init__.py`

7 Cards stream hote hain

Server SSE se ek-ek card bhejta hai. Har card: purani line, nayi line, ek plain reason, aur “kahan se aaya” ka quote.
`POST /api/tailor · server-sent events`

8 Apply — poora browser mein

Apply dabane pe koi server call nahi jaati. Document browser mein hi badalta hai, isliye turant hai aur undo bhi turant. Server sirf download ke waqt aata hai.
`lib/document.ts`

9 Export — usi format mein wapass

SourceAnchors ki wajah se edits tumhari *apni* file mein jaate hain. .docx aur .tex byte-identical wapass aate hain agar kuch na badla ho.
`export/runs.py · export/sources.py`

10 Save — Library mein record ban jaata hai

Job post ka frozen snapshot + jo CV bheja + har change ka log, sab ek saath. Account ki zaroorat nahi; sign in baad mein karo toh kaam tumhare saath aa jaata hai.
`POST /api/records`

AI architecture

03 MODELS AUR PROMPT

Do models, apne-apne kaam ke liye. Mehenga model wahan jahan quality hi product hai; sasta model wahan jahan kaam mechanical hai.

MODEL	KAAM	KYUN
gemini-2.5-flash	Tailoring suggestions, Gap Coach	Yahi asli product hai — judgement chahiye
gemini-3.1-flash-lite	Job post parse, coverage check	Extraction hai, sasta aur tez kaafi hai

EK ZAROORI BAAT

Google ke free tier pe saaf likha hai “data used for training”. Aptly ka input CV hai — naam, phone, address, poori job history. Isliye code startup pe hi paid key assert karta hai aur free tier pe chupke se girne se rokta hai. Chup-chaap free pe girna hi asli khatra hai.

Prompt kaise bana hai

Model ko sirf do cheezein source ke roop mein milti hain: CV ke **editable nodes** (bullets, summary, skill lines), aur **Story Bank** ke items jab woh bane.

Job post *source material nahi hai*. Ye jaan-boojh ke hai — advert employer ki wish list hai, applicant ka itihaas nahi. Agar job post ko source maan lete toh model employer ki maangi hui skills ko chupke se tumhare CV mein daal deta.

Kya editable hai, kya nahi

Naam, employer, job title, dates — ye **facts** hain, prose nahi. Model ko ye editable ke roop mein milte hi nahi. Isse fabrication ki ek poori category prompt tak pahunchne se pehle khatam ho jaati hai: rewrite pass tumhe “Senior” promote nahi kar sakta.

Structured output

Har call Pydantic response_schema ke saath jaati hai, toh model prose nahi de sakta jahan data chahiye. Har suggestion mein **provenance** field mandatory hai — kaunsi line se aaya aur uska exact quote. Provenance na ho toh suggestion tum tak pahunchti hi nahi.

Speed aur cost

	PEHLE	AB
Ek run	117 s	~11 s
Pehla card	~40 s	~7 s
Cost	\$0.070	\$0.015

Teen cheezon se: sections parallel chalte hain, thinking budget capped hai, aur job post ab prompt mein *ek baar* jaata hai — pehle parsed summary aur poora raw advert dono ja rahe the, jisse CV ki 3–4 lines prompt mein dab jaati thin.

No-fabrication validator

04 PRODUCT KA CENTRAL CLAIM

Ye product ka central claim hai, isliye ise prompt pe nahi chhoda gaya. Prompt ek *tendency* deta hai; ye **guarantee** hai — deterministic Python jo model ke har suggestion ko dobara check karta hai.

- | | | |
|---|--|--------|
| 1 | Anchor
Line abhi bhi wahi kehti hai jo suggestion sochti hai? Warna tumne beech mein edit kiya — overwrite nahi hoga. | REJECT |
| 2 | Provenance
Cited source exist karta hai, aur uska quote sach mein usme milta hai? Jo apna kaam dikha na sake, woh nahi jaayega. | REJECT |
| 3 | Claims
Har figure, naam aur technical token pehle se tumhare material mein hona chahiye. Numbers pe sabse sakht — jo digit source mein nahi, woh hard block. | REJECT |
| 4 | Title loss
“Data Science Intern — CSC India:” bullet ke aage se nahi ud sakta. Woh batata hai kaam <i>kahan</i> hua. | REJECT |
| 5 | Deletions
Skills line reorder ho sakti hai, kaat nahi sakti. Tum C++ ab bhi jaante ho, chahe ye advert usko na maange. | REJECT |
| 6 | Stuffing
Koi term score badhane ke liye baar-baar nahi aa sakta. | REJECT |
| 7 | Proportion aur vagueness
Line bahut lambi ho gayi = padding. Ya matlab wale shabd hata ke kuch add na kiya = dhundhla, tighter nahi. | FLAG |
| 8 | Dropped details
Employer ka naam ya koi number line se gayab hua toh tumhe dikhega — faisla tumhara. | FLAG |

Layers 1–6 **reject** karte hain kyunki woh sach naapte hain. 7–8 sirf **flag** karte hain kyunki woh taste naapte hain — aur ek suggestion jo tum dekh ke khud judge kar sako, woh chupke se hataye jaane se behtar hai.

Asli output

Jaan-boojh ke ~20% match wali jodi: frontend developer ka CV, data engineer ki job. JD Airflow, dbt, Kafka, Spark, Snowflake maangta hai — ek bhi CV mein nahi.

```
COVERAGE 2 / 11
  has      SQL, Python
  missing  Airflow, dbt, AWS, Spark, Kafka, Snowflake, BigQuery...

[1] TECHNICAL SKILLS
  Languages: JavaScript, TypeScript, Python, HTML5, CSS3
  → Languages: Python, SQL, JavaScript, TypeScript, HTML5, CSS3

[2] WORK EXPERIENCE
  → Built the internal pricing dashboard end to end, writing SQL...

[3] WORK EXPERIENCE
  → Wrote a Python script for data quality, reconciling the...

accepted=4 rejected=1 {dropped_skill: 1} 10.3s $0.0104
FABRICATED TOOLS: none
```

Dhyan do: model ne skills line se PostgreSQL aur SQL *delete* karne ki koshish ki — validator ne roka. Aur Airflow/dbt/Kafka kahin nahi aaye, jabki JD unhi ko maang raha tha. Yahi asli test hai.

File formats ka jhanjhat

05 FORMAT PRESERVATION

Tumne kaha tha: jis format mein user de, usi mein wapas. Ye teen formats mein **poori tarah** hota hai, aur PDF mein *lagbhag*.

FORMAT	FIDELITY	KAISE
.docx	~95%	Run-level surgery — asli file hi edit hoti hai, styling untouched
.tex	~99%	Source plain text hai — content patch, baaki sab waisa
.txt / .md	100%	Line span replacement
.pdf	~80% visual	Style profile nikaal ke rebuild — edit nahi

PDF EDIT KYUN NAHI HO SAKTA

PDF ek *print* format hai, document format nahi. Usme “paragraph” ya “sentence” hoti hi nahi — sirf “is (x,y) pe ye glyph draw karo”. Text reflow ka concept hai hi nahi. Agar naya bullet purane se chaar shabd lamba hua, toh woh literally next line pe overlap karega ya cut jaayega.

Isliye PDF ke liye: har character ka font, size, colour aur position padh ke **style profile** banaya jaata hai, phir usi style mein naya PDF render hota hai. Original jaisa *dikhta* hai, ATS-clean hota hai, aur aage se sach mein editable ho jaata hai. UI mein saaf likha jaata hai ki rebuild hua.

Do cheezein jo genuinely mushkil thin

Wrapped bullets. Tumhare CV mein har bullet 3–4 lines pe wrap hota tha. Parser har physical line ko alag node maan raha tha, isliye suggestions aadhe vaakyon pe aati thin — “and performed model”. Ab hanging indent detect hota hai, jo full-stops ke aar-paar bhi kaam karta hai.

Bullet glyphs jo hote hi nahi. Ek test PDF mein 5 bullets banaye, sirf 4 glyphs extract hue — subset fonts mein dingbats aksar text layer mein aate hi nahi. Jab glyph missing ho toh do alag bullets aur ek wrapped line bilkul ek jaise dikhte hain.

Tech stack

06 HAR CHOICE KA KAARAN

LAYER	CHOICE	KYUN YAHİ
Frontend	Next.js 16 + React 19	SEO landing + interactive app, ek codebase
Styling	Tailwind v4	Design doc ke exact tokens theme layer mein
3D	React Three Fiber	Landing page ka scroll narrative
Backend	FastAPI + Python 3.12	Async SSE streaming, uploads, Pydantic
Packages	uv	Tumhari requirement · lockfile, reproducible
Database	SQLite → Postgres	Ek hi schema dono pe · Supabase pe sirf URL badlegi
ORM	SQLAlchemy 2.0 async	Portable models, real migrations
Auth	Dev sign-in → Supabase	Ek interface, do implementations
LLM	Gemini (google-genai)	Pydantic response_schema se structured output
DOCX	python-docx	MIT · run-level in-place editing
PDF read	pdfminer.six	MIT, pure Python · per-char font data
PDF write	Typst	Ek self-contained wheel, koi system dependency nahi

Teen library decisions jo tumhari machine ne force kiye

Ye teeno “best practice” nahi hain — ye tumhare Intel Mac (macOS 12.7, 8 GB) pe kaam karne ki wajah se hain, aur teeno verified hain.

- **pdfplumber** → **pdfminer.six** — pdfplumber pypdfium2 laata hai jiske wheels macOS 13+ ke hain. Is machine pe install hi nahi hota.
- **WeasyPrint** → **Typst** — WeasyPrint ko system cairo/pango chahiye, aur Homebrew macOS 12 ke liye bottle ship nahi karta. Matlab ghanon ka source build. Typst ko kuch nahi chahiye.
- **cryptography<49 pinned** — v49+ se upstream ne Intel macOS wheels band kar diye, sirf Apple Silicon bache.

Ek aur: **PyMuPDF jaan-boojh ke avoid kiya**. Sabse fast hai par AGPL v3 — commercial product mein ya toh poora source open karna padta ya licence khareedni padti.

Kya bana, kya baaki

07 STATUS

Chal raha hai

FEATURE	KYA KARTA HAI
Tailor loop	Two-box drop, streaming change cards, one-tap apply, undo, apply-all
Live preview	CV side mein, badli hui line amber flash karke settle hoti hai
Coverage meter	Kaunse job terms cover hain, kaunse missing — imaandari se
Format preserve	5 formats, .docx/.tex byte-identical wapas
Validator	8 layers, har suggestion pe
Library	Frozen job post + jo CV bheja + change log · search, status, notes
Anonymous first	Bina account sab kuch · sign in pe kaam saath aata hai
Erase everything	Sach mein sab delete, kuch peeche nahi
Landing page	Scroll narrative, 3D paper jo har section ke saath badalta hai

Aage ka

FEATURE	KYUN MATTER KARTA HAI
Story Bank	Har achievement ek baar likho, tagged. Tailoring usse draw kare
Recruiter-Ready Card	Product ka signature — call ke liye ek screen
Gap Coach	Har missing requirement: bridge ho sakta hai, ya asli gap hai
Interview questions	Job post se predicted, answers Story Bank se
Version history	Har job ke liye kya badla, rollback
Browser clipper	Ek click mein job post seedha record mein
Payments	Tiers aur pricing page ban chuke, processing baaki

RAG KAHAN FIT HOTA HAI

Core tailoring loop mein **nahi**. Ek CV 500 shabd ka hai, poora prompt mein aa jaata hai — retrieve karne ko kuch nahi. Ulta validator ko poora CV chahiye, aur agar retrieval aadha chhod de toh woh sach ko bhi “invented” bolne lagega.

Story Bank mein bilkul fit hota hai. Jab 200 achievements likhe honge tab sab prompt mein nahi bhej sakte — is job ke liye top 10 relevant nikalna exactly retrieval ka kaam hai. Yaani tumhari instinct sahi thi, bas timing aage hai.

Chalane ka tarika

08 SETUP

Do terminal:

```
# backend – http://localhost:8000
uv sync
uv run uvicorn aptly.main:app --reload --port 8000

# frontend – http://localhost:3000
cd frontend && npm run dev
```

.env mein sirf GEMINI_API_KEY chahiye. Database apne aap ban jaata hai — SQLite file, koi setup nahi.

Jab kuch galat lage

```
grep -E "tailor.section|tailor.rejected" /tmp/aptly-api.log | tail -20
```

tailor.section batata hai har section ne kitne suggestions diye — *zero bhi*. Aur tailor.rejected batata hai kya discard hua aur kyun. Yahi do lines project ke aadhe bugs pakad chuki hain.

Folder structure

```
aptly/
├── backend/    FastAPI – parsers, LLM, validator, API + tests
├── frontend/   Next.js – landing, Tailor, Library
├── docs/       design doc + demo CV/JD pair
└── .env        tumhari key
```

Imaandari se

09 **KYA ABHI NAHI HAI**

Kuch cheezein jo is doc mein saaf honi chahiye, warna ye brochure ban jaayega.

Dev sign-in asli authentication nahi hai

Email leta hai, password nahi. `APPLY_ENV=production` pe chalne se saaf mana kar deta hai. Uska kaam sirf itna tha ki Supabase aane se pehle Library ban aur test ho sake.

Tenancy app code mein enforce hoti hai

SQLite mein row-level security hoti hi nahi. Har query owner se filter karti hai, aur test hai ki ek account doosre ka data na dekh sake. Par Supabase pe RLS policies *bhi* lagani chahiye — dusri line of defence ke roop mein.

PDF rebuild byte-identical nahi hai

~80% visual match hai. Achha hai, par agar tumhara CV bahut design-heavy hai toh .docx dena behtar rahega.

Payments abhi nahi hain

Pricing page aur tiers ban chuke hain, entitlements table ready hai — par koi checkout nahi. Page pe bhi yahi likha hai, kyunki ek “Subscribe” button jo chupke se kuch na kare, uska hona na hone se bura hai.

Distribution alag kaam hai

Main product quality, speed aur trust bana sakta hoon. Log aayenge ya pay karenge — uski guarantee nahi de sakta. SEO, Reddit, Product Hunt, LinkedIn — uska plan alag se banega.

Sab kuch verified hai: 202 backend tests pass, lint clean, TypeScript clean, aur tailoring loop asli Gemini API ke against chal ke check kiya gaya.